VPC Challenge-4

Use Case: Setting up Transit Gateway and VPC Endpoints for a Multi-VPC Architecture

Scenario:

A large organization is migrating its on-premises infrastructure to the AWS cloud. The organization's architecture involves multiple VPCs for different departments and applications, each requiring secure communication with centralized services and external resources.

The IT team needs to design and implement a scalable and efficient network architecture to accommodate the organization's growth and ensure robust connectivity between VPCs and external services.

Objectives:

1. Design and deploy a scalable network architecture using AWS Transit Gateway to simplify network connectivity between multiple VPCs.

Step 1:-Create 3 VPC, 3 subnets, 3 Route tables, 1 IGW.

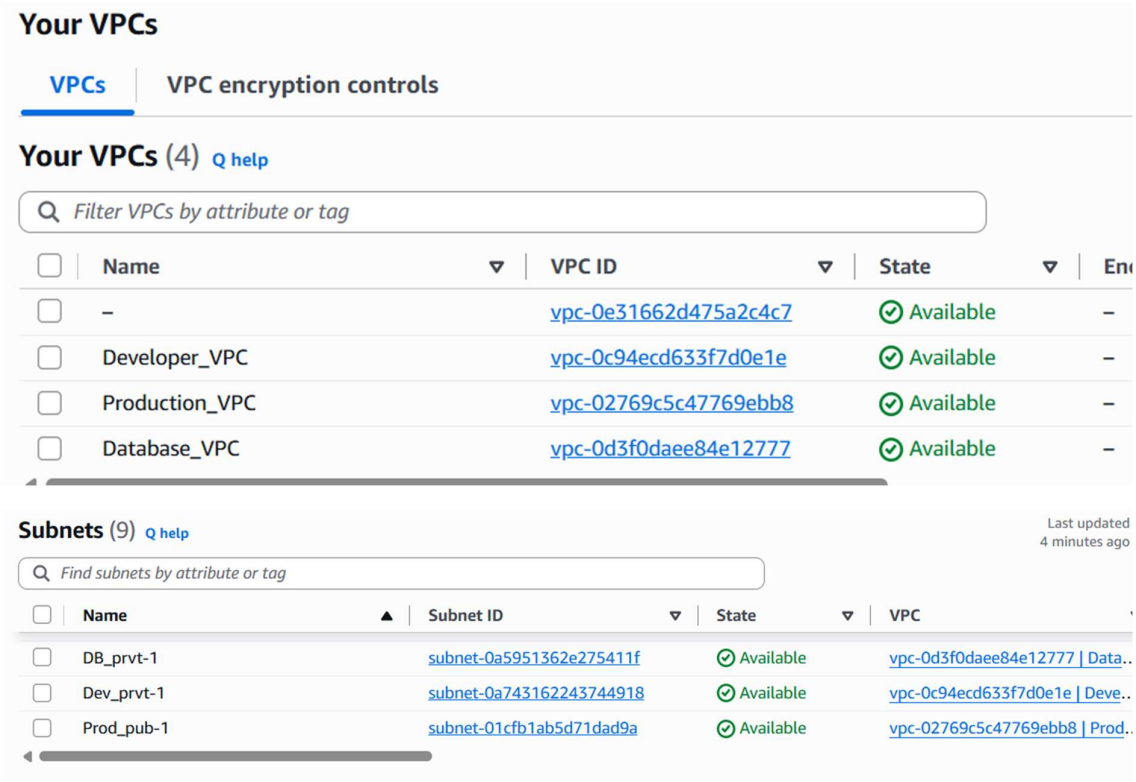
Step 2:- Now go to transit gateway and create it with default Amazon side Autonomous System Number (ASN).

Step 3:- Then create transit gateway attachments with every VPC id.

Step 4:- Then configure this TGW into the route tables of every VPC routes.

Step 5:- Now create 3 instances one public and two private.

Step 6:- Now we can check the connectivity for TGW with ping command.



Route tables (5) Q help

Last upd
3 minutes

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associ...	Edge associations
DB_prvt-RT	rtb-04bb5f7deaf916af9	subnet-0a5951362e2754...	–
Dev_prvt-RT	rtb-0ef4d7c519ac908b0	subnet-0a743162243744...	–
Prod_pub-RT	rtb-05ebfb32d56945828	subnet-01cfb1ab5d71da...	–

Internet gateways (2) Q help

Actions

Find internet gateways by attribute or tag

Name	Internet gateway ID	State	VPC ID
Prod-IGW	igw-0908b0fbb2ff017a3	Attached	vpc-02769c5c47769ebb8 | Production_...
–	igw-0d869fea4975d63e8	Attached	vpc-0e31662d475a2c4c7

Transit gateways (1) Info

Actions

Find transit gateway by attribute or tag

Name	Transit gateway ID	Owner ID	State
TGW Hub	tgw-031891d2dacfe6712	472567279649	Available

Transit gateway attachments (3) Info

Actions | Create transit gateway attachment

Find transit gateway attachment by attribute or tag

1

Name	Transit gateway attachment ID	Transit gateway ID	State	Resource ...	Resource ID
DB_TGW	tgw-attach-02bf1809491351820	tgw-031891d2dacfe6712	Available	VPC	vpc-0d3f0daee84e12777
Dev_TGW	tgw-attach-0a5233c4726663254	tgw-031891d2dacfe6712	Available	VPC	vpc-0c94ecd633f7d0e1e
Prod_TGW	tgw-attach-0d021d51b562f9df3	tgw-031891d2dacfe6712	Available	VPC	vpc-02769c5c47769ebb8

rtb-04bb5f7deaf916af9 / DB_prvt-RT

Details | Routes | Subnet associations | Edge associations | Route propagation | Tags | Related resources

Routes (3)

IP version: Both | Edit routes

Filter routes

1

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	tgw-031891d2dacfe6712	Active	No	Create Route
172.21.0.0/16	tgw-031891d2dacfe6712	Active	No	Create Route
194.69.0.0/16	local	Active	No	Create Route Table

rtb-0ef4d7c519ac908b0 / Dev_prvt-RT

Details | Routes | Subnet associations | Edge associations | Route propagation | Tags | Related resources

Routes (3)

IP version: Both | Edit routes

Filter routes

1

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	tgw-031891d2dacfe6712	Active	No	Create Route
172.21.0.0/16	local	Active	No	Create Route Table
194.69.0.0/16	tgw-031891d2dacfe6712	Active	No	Create Route

rtb-05ebfb32d56945828 / Prod_pub-RT

Details | Routes | Subnet associations | Edge associations | Route propagation | Tags | Related resources

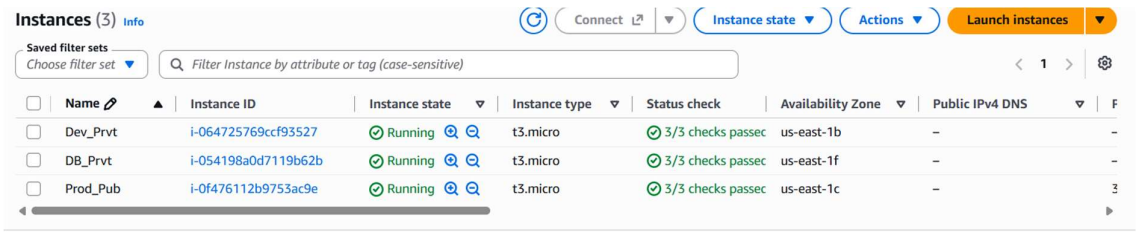
Routes (4)

IP version: Both | Edit routes

Filter routes

1

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0908b0fbb2ff017a3	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table
172.21.0.0/16	tgw-031891d2dacfe6712	Active	No	Create Route
194.69.0.0/16	tgw-031891d2dacfe6712	Active	No	Create Route




```
Faisal@LAPTOP-4DO8SS4E UCRT64 ~/Downloads
$ ssh -i EC2.pem ec2-user@35.172.200.84
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Tue Nov  3 08:58:27 2026 from 103.203.172.230
[ec2-user@ip-10-0-97-66 ~]$ ping 194.69.0.152
PING 194.69.0.152 (194.69.0.152) 56(84) bytes of data.
^C
--- 194.69.0.152 ping statistics ---
379 packets transmitted, 0 received, 100% packet loss, time 393148ms

[ec2-user@ip-10-0-97-66 ~]$ ping 194.69.0.152
PING 194.69.0.152 (194.69.0.152) 56(84) bytes of data.
64 bytes from 194.69.0.152: icmp_seq=192 ttl=126 time=2.37 ms
64 bytes from 194.69.0.152: icmp_seq=193 ttl=126 time=0.933 ms
64 bytes from 194.69.0.152: icmp_seq=194 ttl=126 time=0.837 ms
64 bytes from 194.69.0.152: icmp_seq=195 ttl=126 time=0.853 ms
64 bytes from 194.69.0.152: icmp_seq=196 ttl=126 time=0.888 ms
```

```
[ec2-user@ip-10-0-97-66 ~]$ ping 172.21.14.175
PING 172.21.14.175 (172.21.14.175) 56(84) bytes of data.
64 bytes from 172.21.14.175: icmp_seq=1 ttl=126 time=4.12 ms
64 bytes from 172.21.14.175: icmp_seq=2 ttl=126 time=0.894 ms
64 bytes from 172.21.14.175: icmp_seq=3 ttl=126 time=0.802 ms
64 bytes from 172.21.14.175: icmp_seq=4 ttl=126 time=0.834 ms
64 bytes from 172.21.14.175: icmp_seq=5 ttl=126 time=0.858 ms
```

2. Configure VPC endpoints to securely access AWS services without internet gateways or NAT gateways, ensuring data privacy and minimizing exposure to external threats.

Step 1:- First create a IAM user with s3 full access policy. Then take access key and secret key from the user.

Step 2:- Now create a S3 bucket.

Step 3:- Go to VPC service and select Endpoint then create new endpoint disable the DNS configurations, We need new security group for this Endpoint.

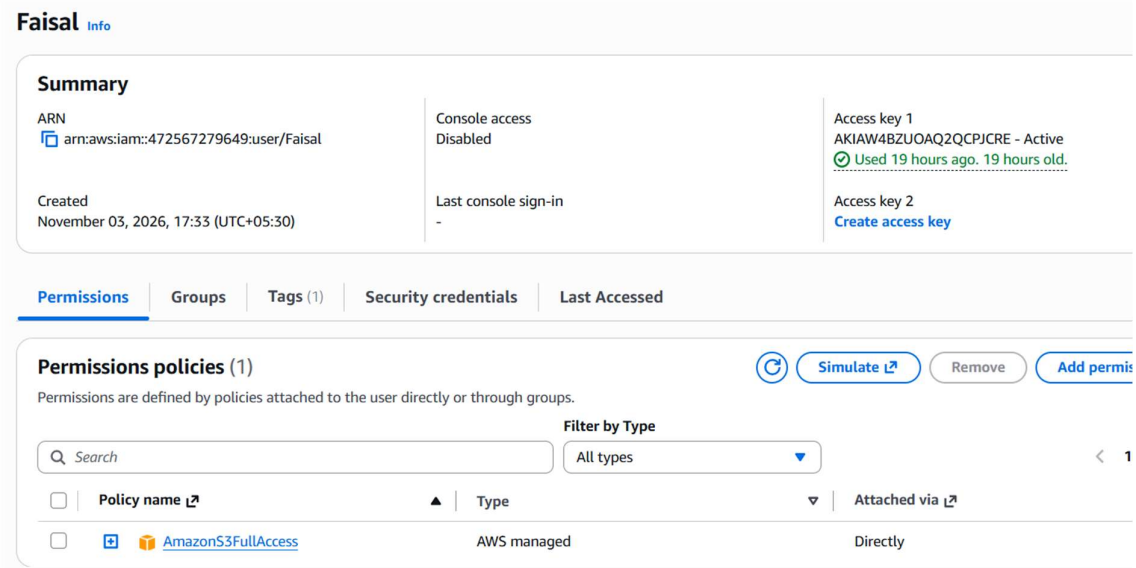
Step 4:- In this security configure all VPC ip into HTTPS port.

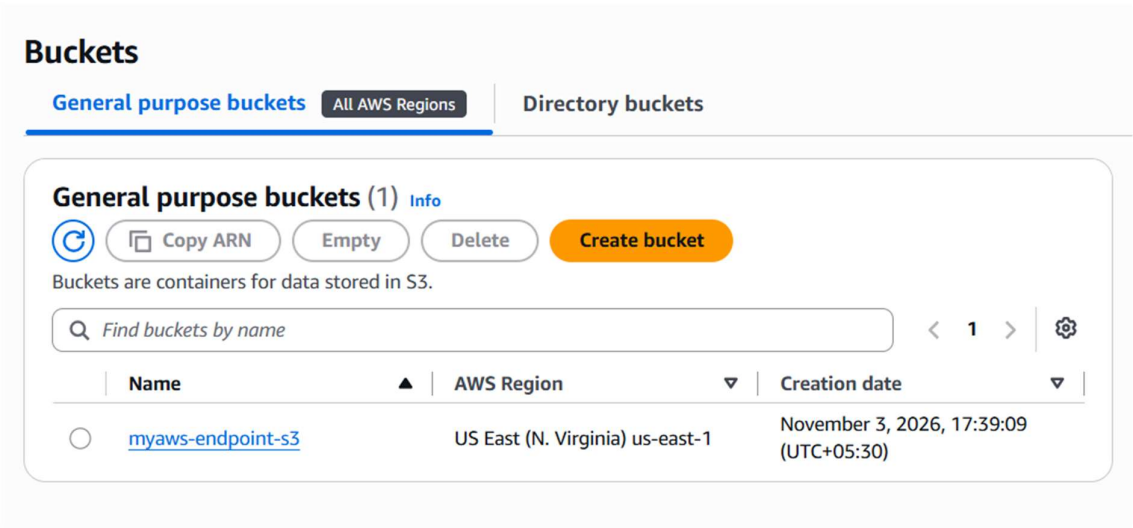
Step 5:- Now select ssm service in the Services while creating the Endpoint. Then add the VPC where we want to attach the Endpoint.

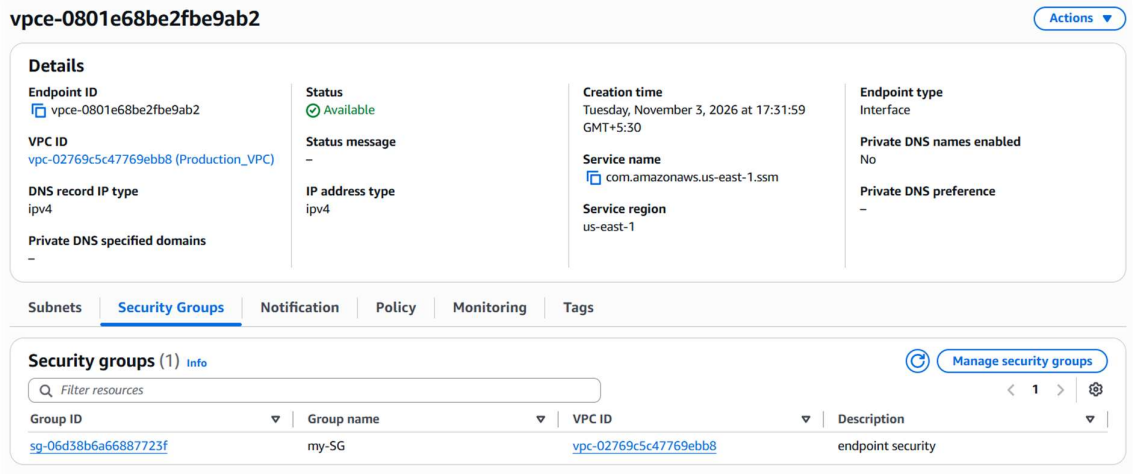
Step 6:- Connect with instance, then connect to private instance.

Step 7:- Use aws configure. Add the credentials there.

Step 8:- Then use aws s3 ls to see S3 bucket name.







sg-06d38b6a66887723f - my-SG

Actions

Details

Security group name	Security group ID	Description	VPC ID
my-SG	sg-06d38b6a66887723f	endpoint security	vpc-02769c5c47769ebb8

Owner	Inbound rules count	Outbound rules count
472567279649	3 Permission entries	1 Permission entry

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - *new* | Tags

Inbound rules (3)

Manage tags | Edit inbound rules

Search

1

Security group rule ID	IP version	Type	Protocol	Port range	Source
sgr-0429b2eb53c0cea05	IPv4	HTTPS	TCP	443	10.0.0.0/16
sgr-09b4a4860eb2b667d	IPv4	HTTPS	TCP	443	172.21.0.0/16
sgr-06dc69c5d6ab1d18d	IPv4	HTTPS	TCP	443	194.69.0.0/16

Instances (3) Info

Connect | Instance state | Actions | Launch instances

Saved filter sets: Choose filter set

Filter Instance by attribute or tag (case-sensitive)

1

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
Dev_Prvt	i-064725769ccf93527	Running	t3.micro	3/3 checks passec	us-east-1b	–
DB_Prvt	i-054198a0d7119b62b	Running	t3.micro	3/3 checks passec	us-east-1f	–
Prod_Pub	i-0f476112b9753ac9e	Running	t3.micro	3/3 checks passec	us-east-1c	–

```
[ec2-user@ip-10-0-97-66 ~]$ aws --version
aws-cli/2.38.0 Python/3.9.25 Linux/6.18.56-125.164.amzn2023.x86_64 source/x86_64.amzn
[ec2-user@ip-10-0-97-66 ~]$ aws configure

Tip: You can deliver temporary credentials to the AWS CLI using your AWS Console sess

AWS Access Key ID [None]: AKIAW4BZUOAQ2QCPJCRE
AWS Secret Access Key [None]: W6nAjQEiTm7oENE6YxOcIfCtplQKdgPKmunV8HOP
Default region name [None]: us-east-1
Default output format [None]: json
[ec2-user@ip-10-0-97-66 ~]$ aws s3 ls

aws: [ERROR]: An error occurred (AccessDenied) when calling the ListBuckets operation
ot authorized to perform: s3:ListAllMyBuckets because no identity-based policy allows
[ec2-user@ip-10-0-97-66 ~]$ aws s3 ls

aws: [ERROR]: An error occurred (AccessDenied) when calling the ListBuckets operation
ot authorized to perform: s3:ListAllMyBuckets because no identity-based policy allows
[ec2-user@ip-10-0-97-66 ~]$ aws s3 ls
2026-11-03 12:09:09 myaws-endpoint-s3
[ec2-user@ip-10-0-97-66 ~]$
```